SAMPLE

Light

The Fastest Thing in the Universe

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Introduction

Light, visible light, or visible radiation is electromagnetic radiation that can be perceived by the human eye. Visible light spans the visible spectrum and is usually defined as having wavelengths in the range of 400-700 nanometers (nm), corresponding to frequencies of 750-420 terahertz. The visible band sits adjacent to the infrared (with longer wavelengths and lower frequencies) and the ultraviolet (with shorter wavelengths and higher frequencies), called collectively optical radiation.

In physics, the term "light" may refer more broadly to electromagnetic radiation of any wavelength, whether visible or not. In this sense, gamma rays, X-rays, microwaves and radio waves are also light. This is why visible radiation is commonly termed visible light. The primary properties of light are intensity, propagation direction, frequency or wavelength spectrum, and polarization. Its speed in vacuum, 299792458 m/s, is one of the fundamental constants of nature. All electromagnetic radiation exhibits some properties of both particles and waves.

Single, massless elementary particles, or quanta, of light called photons can be detected with specialized equipment; phenomena like interference are described by waves. Most everyday interactions with light can be understood using geometrical optics. Quantum optics is an important research area in modern physics. The main source of natural light on Earth is the Sun. Historically, another important source of light for humans has been fire, from ancient campfires to modern kerosene lamps.

With the development of electric lights and power systems, electric lighting has effectively replaced firelight. Generally, electromagnetic radiation (EMR) is classified by wavelength into radio waves, microwaves, infrared, the visible spectrum that we perceive as light, ultraviolet, X-rays and gamma rays. The designation "radiation" excludes static electric, magnetic and near fields. The behavior of EMR depends on its wavelength. Higher frequencies have shorter wavelengths and lower frequencies have longer wavelengths.

When EMR interacts with single atoms and molecules, its behavior depends on the amount of energy per quantum it carries. EMR in the visible light region consists of quanta (called photons) that are at the lower end of the energies that are capable of causing electronic excitation within molecules, which leads to changes in the bonding or chemistry of the molecule.

How light comes into existence is a story with several chapters of its own. Hot matter radiates it, as any flame or filament shows. Atoms and molecules give it off in discrete colours when their electrons fall from excited states, the principle behind fireworks, neon signs and the aurora. And a handful of more exotic mechanisms complete the list: electroluminescence, the direct conversion of electricity to light in LEDs; scintillation, the faint flash a charged particle leaves in certain crystals; sonoluminescence, light from collapsing bubbles in a liquid; triboluminescence, the sparks of light that wintergreen candies give off when crushed; and, at the highest energies, the annihilation of matter with antimatter and the decay of radioactive nuclei.

Light is measured with two main alternative sets of units: radiometry consists of measurements of light power at all wavelengths, while photometry measures light with wavelength weighted with respect to a standardized model of human brightness perception.

Chapter 1: The Discovery of Light

The oldest surviving optical writings are Indian and Greek. Around the fifth century BC, Empedocles taught that light must take time to travel, and Greek atomists supposed that images peeled off objects and streamed into the eye, a theory of vision that proved remarkably durable. In ancient India, the schools of Samkhya and Vaisheshika treated light as one of the fundamental elements, and Indian atomists later described it as a stream of minute particles moving too fast for the eye to resolve one by one. Whatever the reasoning behind them, such guesses put the question that the next two thousand years would spend answering: does light move, and if so, how fast?

The ancients' question reached its answer in the century of telescopes. Galileo tried the experiment himself, standing lantern-holders on distant hills and asking them to uncover at a flash, and concluded honestly that he could not tell whether light was instantaneous or merely very fast; the gaps he was trying to measure were microseconds, and his clock was a human hand. The answer needed an object farther away than any hill, and it arrived with the moons of Jupiter, the first genuinely cosmic clock in the history of measurement.

In 1021, Alhazen (Ibn al-Haytham) published the Book of Optics, in which he presented a series of arguments dismissing the emission theory of vision in favour of the now accepted intromission theory, in which light moves

from an object into the eye. This led Alhazen to propose that light must have a finite speed, and that the speed of light is variable, decreasing in denser bodies.

Alhazen argued that light is substantial, and that its propagation takes time even if the senses hide this from us. Also in the 11th century, Ab? Rayh?n al-B?r?n? agreed that light has a finite speed, and observed that the speed of light is much faster than the speed of sound. In the 13th century, Roger Bacon argued that the speed of light in air was not infinite, using philosophical arguments backed by the writing of Alhazen and Aristotle.

In the 1270s, Witelo considered the possibility of light travelling at infinite speed in vacuum, but slowing down in denser bodies. In the early 17th century, Johannes Kepler believed that the speed of light was infinite since empty space presents no obstacle to it. René Descartes argued that if the speed of light were to be finite, the Sun, Earth, and Moon would be noticeably out of alignment during a lunar eclipse.

Although this argument fails when aberration of light is taken into account, the latter was not recognized until the following century. Since such misalignment had not been observed, Descartes concluded the speed of light was infinite. Descartes speculated that if the speed of light were found to be finite, his whole system of philosophy might be demolished. Despite this, in his derivation of Snell's law, Descartes assumed that some kind of motion associated with light was faster in denser media.

Pierre de Fermat derived Snell's law using the opposing assumption, the denser the medium the slower light travelled. Fermat also argued in support of a finite speed of light. In 1629, Isaac Beeckman proposed an experiment in which a person observes the flash of a cannon reflecting off a mirror about one mile (1.6 km) away.

In 1638, Galileo Galilei proposed an experiment, with an apparent claim to having performed it some years earlier, to measure the speed of light by observing the delay between uncovering a lantern and its perception some distance away. He was unable to distinguish whether light travel was instantaneous or not, but concluded that if it were not, it must nevertheless be extraordinarily rapid. According to Galileo, the lanterns he used were "at a short distance, less than a mile".

Assuming the distance was not too much shorter than a mile, and that "about a thirtieth of a second is the minimum time interval distinguishable by the unaided eye", Boyer notes that Galileo's experiment could at best be said to have established a lower limit of about 60 miles per second for the velocity of light. In 1667, the Accademia del Cimento of Florence reported that it had performed Galileo's experiment, with the lanterns separated by about one mile, but no delay was observed.

The actual delay in this experiment would have been about 11 microseconds. The first quantitative estimate of the speed of light was made in 1676 by Ole Rømer. From the observation that the periods of Jupiter's innermost moon Io appeared to be shorter when the Earth was approaching Jupiter than when receding from it, he concluded that light travels at a finite speed, and estimated that it takes light 22 minutes to cross the diameter of Earth's orbit.

Christiaan Huygens combined this estimate with an estimate for the diameter of the Earth's orbit to obtain an estimate of speed of light of 220000 km/s, which is 27% lower than the actual value. In his 1704 book Opticks, Isaac Newton reported Rømer's calculations of the finite speed of light and gave a value of "seven or eight minutes" for light's crossing time between the Sun and us (the modern value is 8 minutes 19 seconds).

Newton queried whether Rømer's eclipse shadows were coloured. Hearing that they were not, he concluded the different colours travelled at the same speed..